

Introduction

- Lack of transparency of machine learning models has raised the question of whether these “black-box” models can be trusted
- Existing Models:
 - Post-Hoc**
 - Aim to explain the prediction after it has been made
 - Doesn't reliably or faithfully represent model's decision-making process
 - Concept Bottleneck Models**
 - Map each input image to a feature vector
 - Accuracy hampered by using linear classifier
- Information Pursuit* (IP) addresses these issues by “playing” a Twenty Questions game in which very few concepts are queried, until a reliable prediction can be made
- Implementing IP requires three ingredients

Querier

- Selects which concepts to query and in what order

Answerer

- Predicts whether a concept selected by the querier is present in the image

Classifier

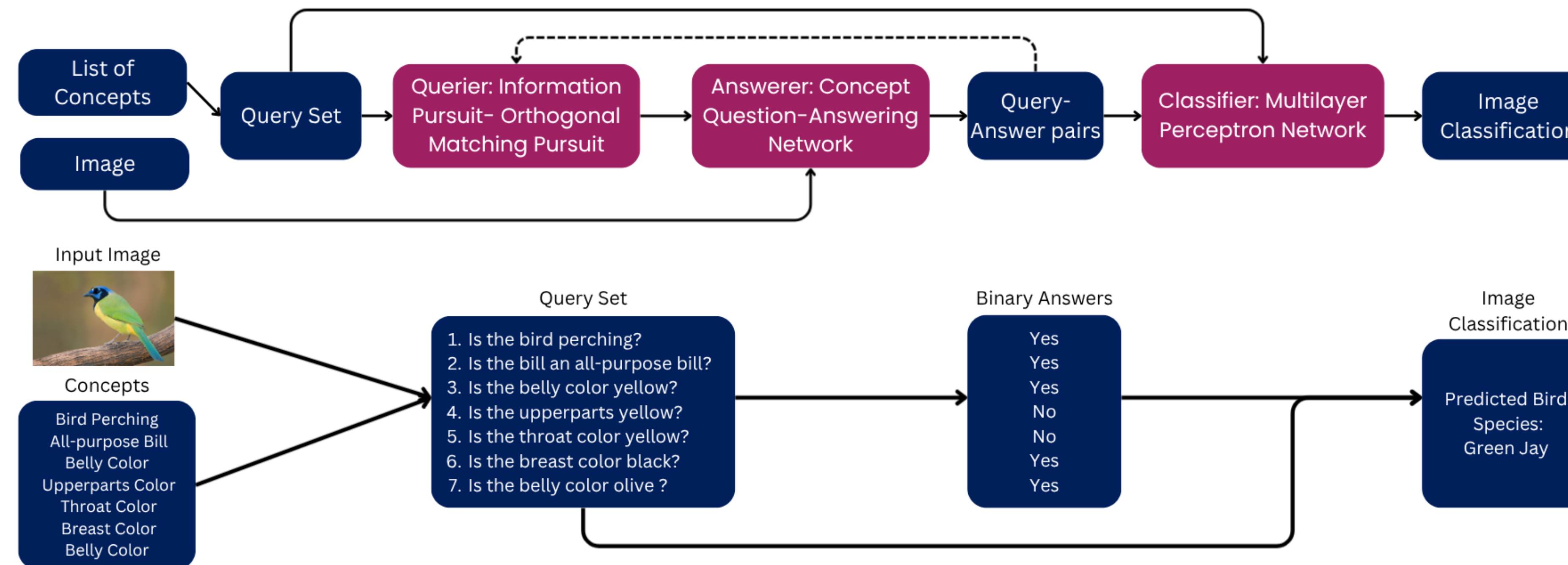
- Predicts the class from the sequence of query-answer pairs

Current Methods vs. IP-OMP-ConceptQA

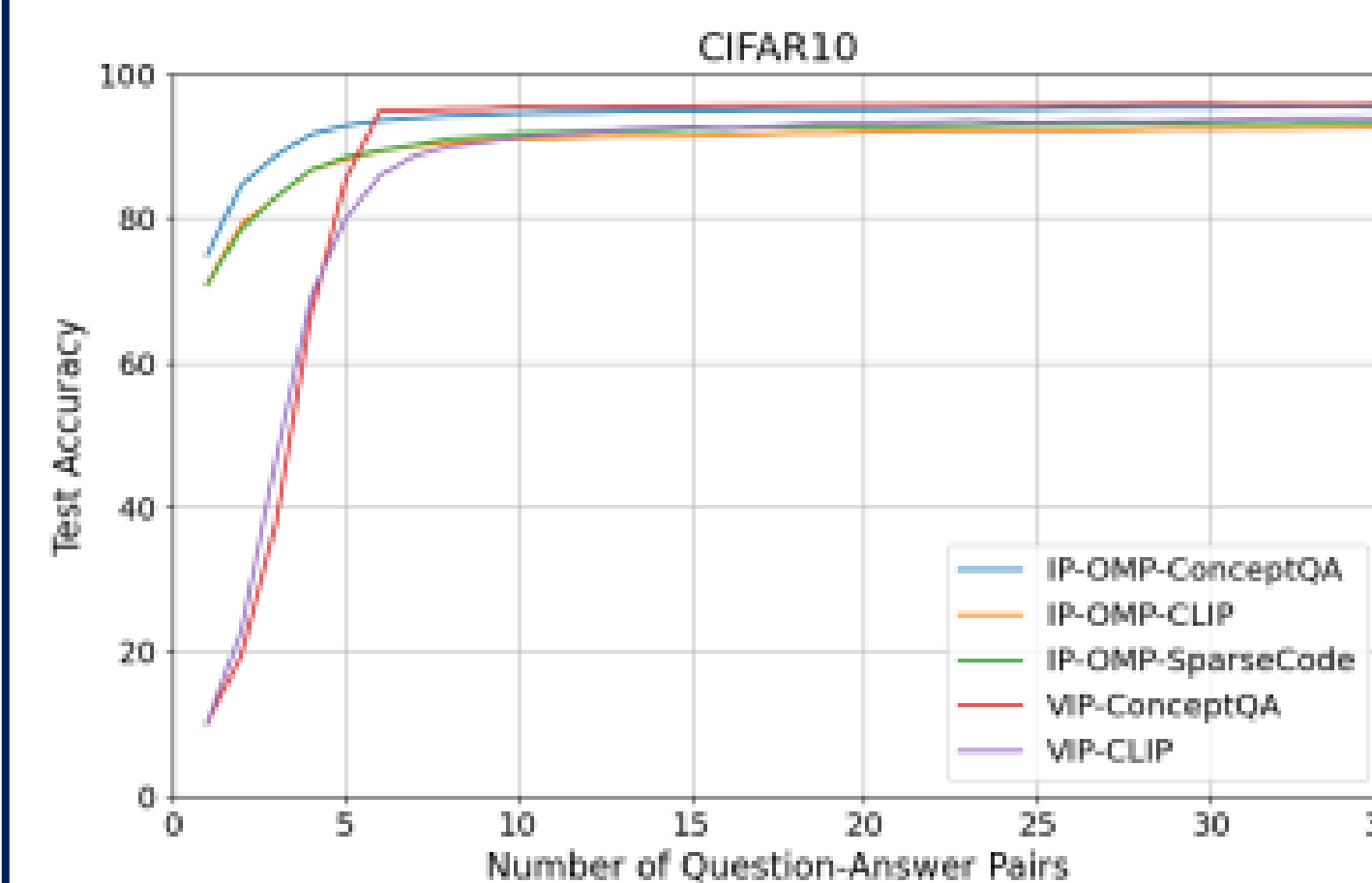
Current Approaches	Efficient Query Selection?	Accurate Zero-Shot Answers?
Concept Bottleneck Models (CBMs)	No selection	Not zero-shot
Information Pursuit (IP)	No	Not zero-shot
Variational Information Pursuit (VIP)	Somewhat	Not zero-shot
Variational Information Pursuit - Concept Question Answering Network (VIP-Concept QA)	Somewhat	Yes
Information Pursuit - Orthogonal Matching Pursuit (IP-OMP)	Yes	No
Information Pursuit- Orthogonal Matching Pursuit- Concept Question Answering Network (IP-OMP-Concept QA)	Yes	Yes

*Aditya Chattopadhyay, Stewart Slocum, Benjamin D. Haeffele, Rene Vidal, and Donald Geman. Interpretable by design: Learning predictors by composing interpretable 267queries. IEEE Transactions on Pattern Analysis and Machine Intelligence, 2022.

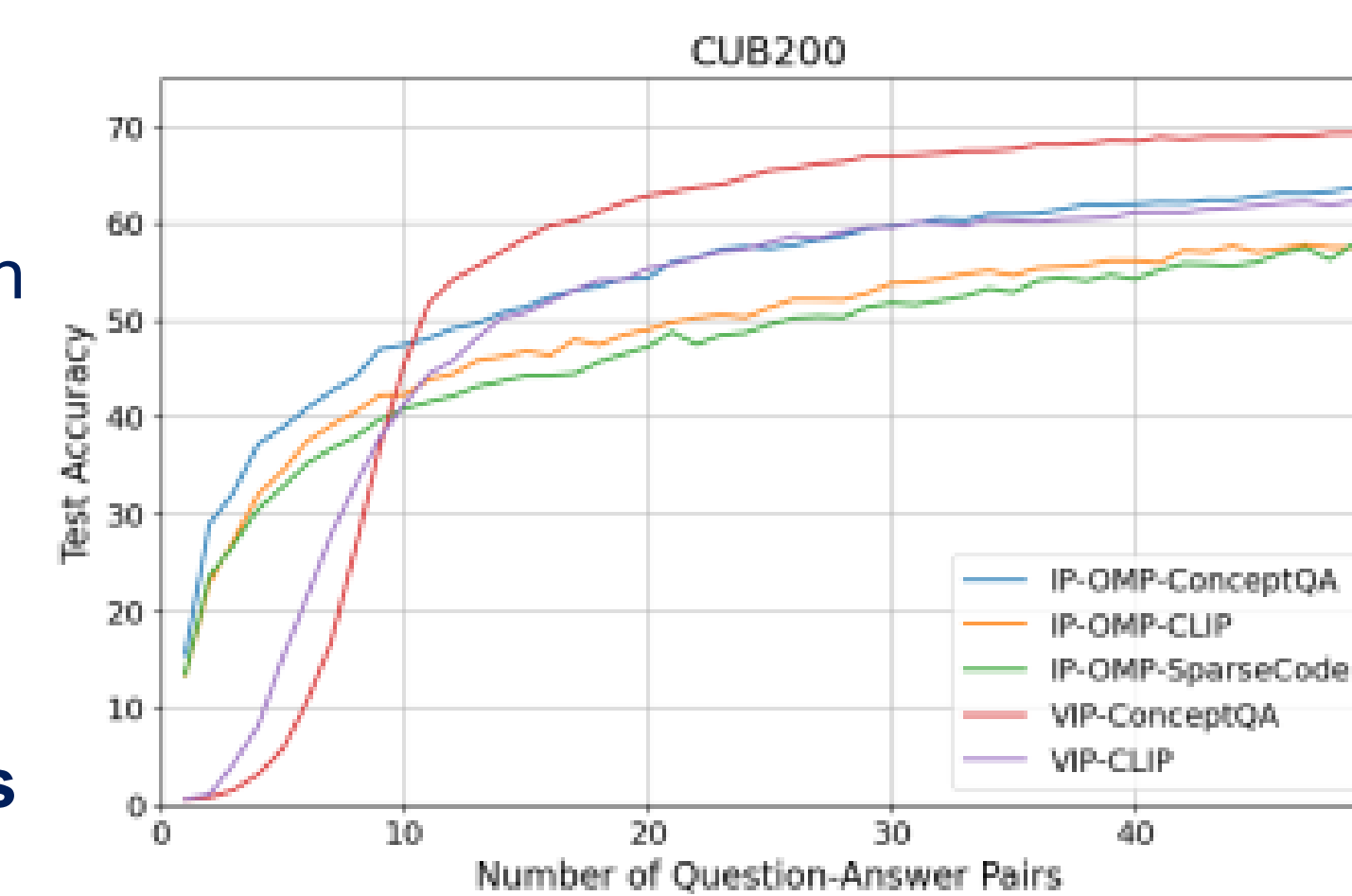
Our Method: IP-OMP-Concept QA



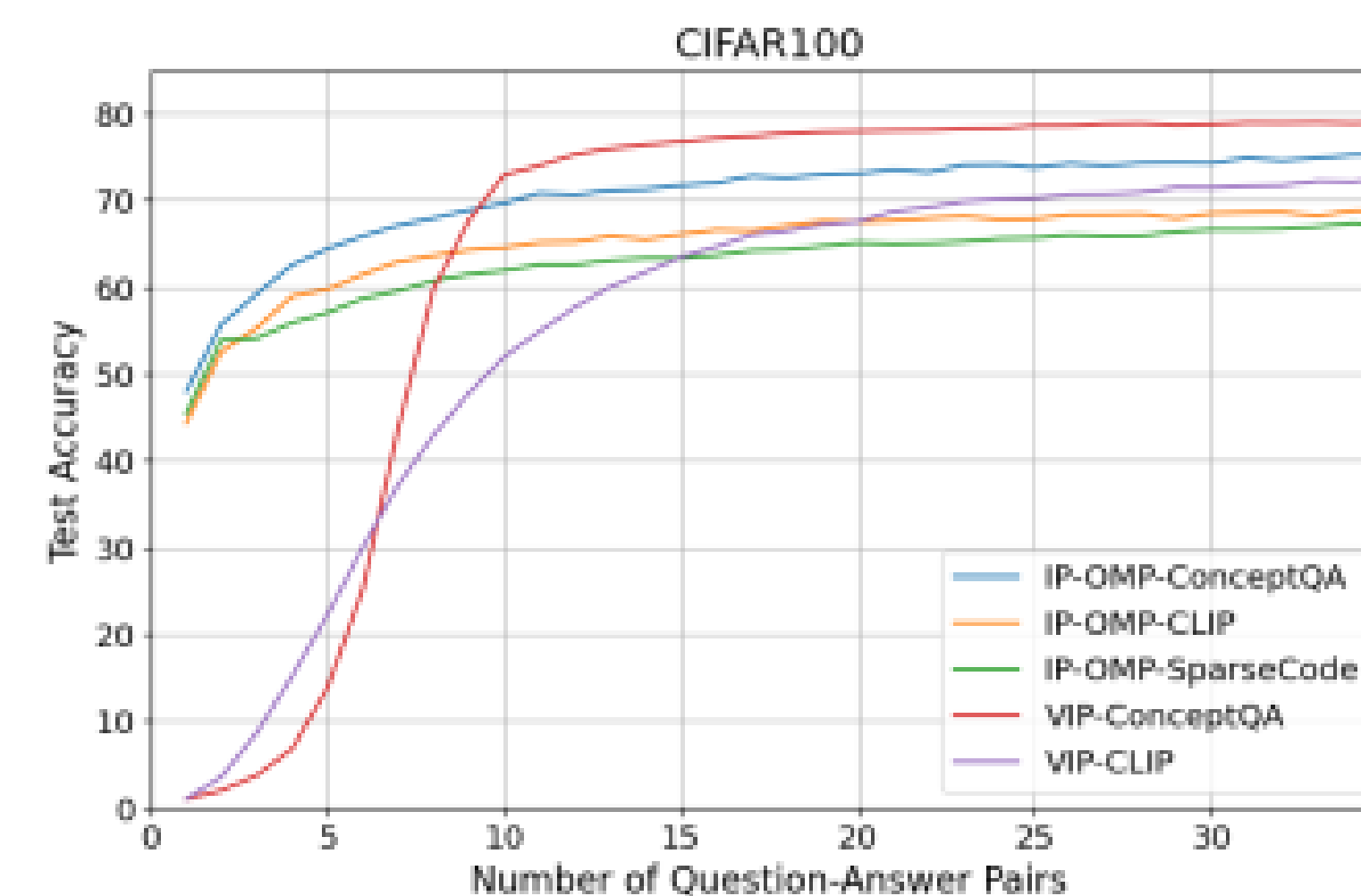
Findings



- IP-OMP-ConceptQA **outperforms** all other algorithms up to a length of **nine**
- For a length **greater than nine**, IP-OMP-Concept QA is **competitive** with VIP-Concept QA, with a **less than 4%** difference in accuracy levels

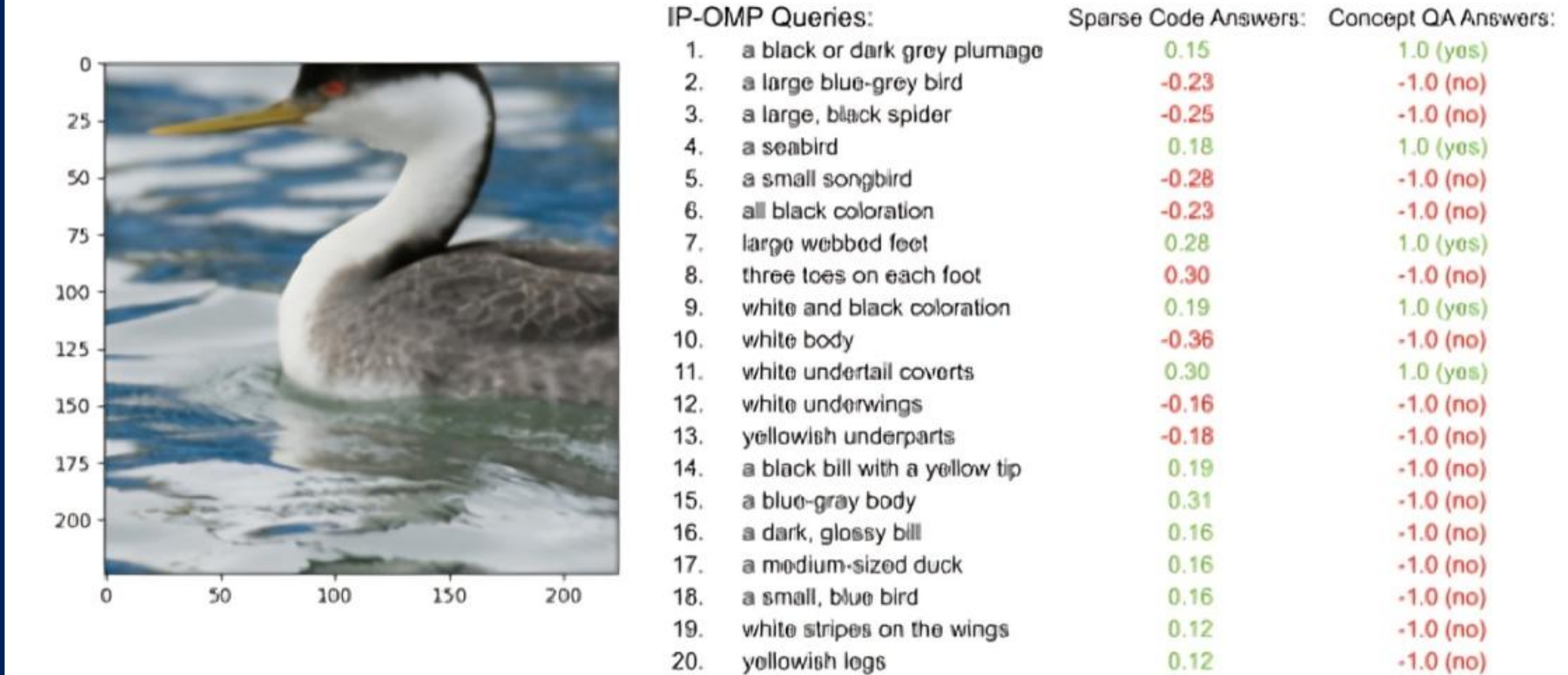


- IP-OMP-ConceptQA achieves the **highest accuracy** for explanation lengths up to **six**
- IP-OMP-ConceptQA is **competitive** with VIP-ConceptQA for an explanation length **greater than six**



- IP-OMP-ConceptQA performs the **best** against all algorithms up to **eleven** explanation length
- IP-OMP-Concept QA is **competitive** to VIP-Concept QA, with a **less than 6%** difference in accuracy levels

Qualitative Evaluation of Explanations



- A positive coefficient (yes, the query is present) is shown in green and a negative coefficient (no, the query is not present) is shown in red.
- Sparse coefficients (answers from IP-OMP) are between (-0.3,0.3), making it uninterpretable
- Concept QA provides binary answers in $\{-1,1\}$, which are interpretable

Conclusion

- Developed IP-OMP-ConceptQA
 - Interpretable-by-design method
 - Efficient query selection method (querier)
 - Accurate zero-shot query answering system (answerer)
 - Produces an interpretable representation (sequence of query-answer pairs) for classification
- IP-OMP-ConceptQA outperforms existing methods in terms of accuracy, interpretability, faithfulness, and efficiency when very short explanations are desired
- IP-OMP is better than VIP at selecting queries when the explanation length is short
- VIP is better than IP-OMP for long explanation lengths